

Lecture 5 — Intro to DC/DC

1 First Averaged Circuit Rules

KCL

$$\begin{aligned}\sum_j i_j &= 0 \\ \frac{1}{T} \int_T \sum_j i_j dt & \\ \sum_j \frac{1}{T} \int_T i_j dt & \\ \sum_j \langle i_j \rangle &= 0\end{aligned}$$

KCL applies to time averaged currents (constant charge).

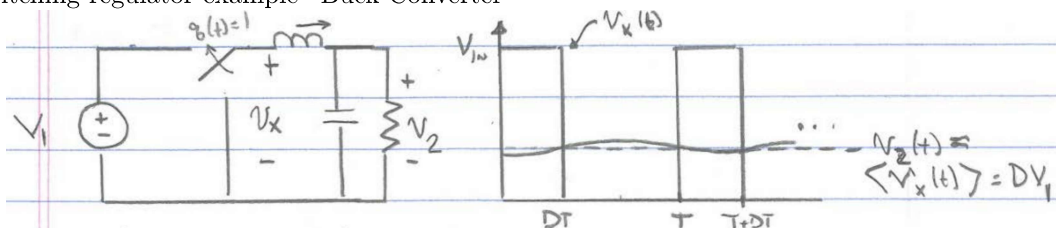
The same is true for KVL ($\sum_k \langle V_k \rangle = 0$).

So for a power converter in periodic steady state:

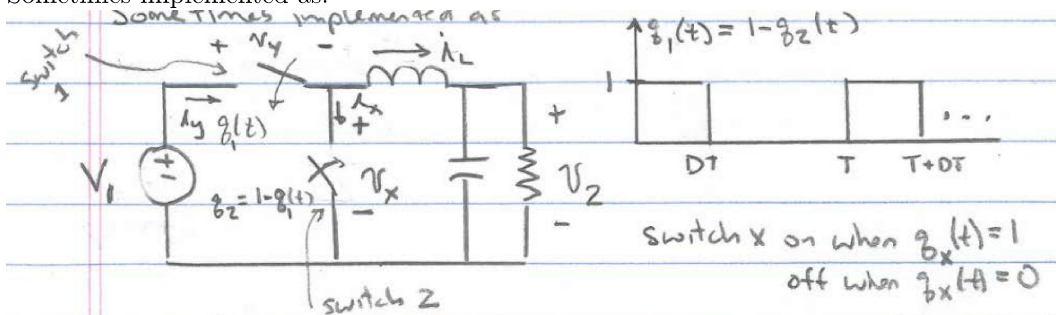
1. Averaged KCL $\sum_j \langle i_j \rangle = 0$
2. Averaged KVL $\sum_k \langle V_k \rangle = 0$
3. Capacitor in P.S.S. $\langle \lambda_c \rangle = 0$
4. Inductor in P.S.S. $\langle V_L \rangle = 0$
5. If system lossless (Conservation of energy) $\langle P_{in} \rangle = \langle P_{out} \rangle$

2 Review

Switching regulator example “Buck Converter”



Sometimes implemented as:



Text on bottom: Switch X on when $q_x(t) = 1$, off when $q_x(t) = 0$

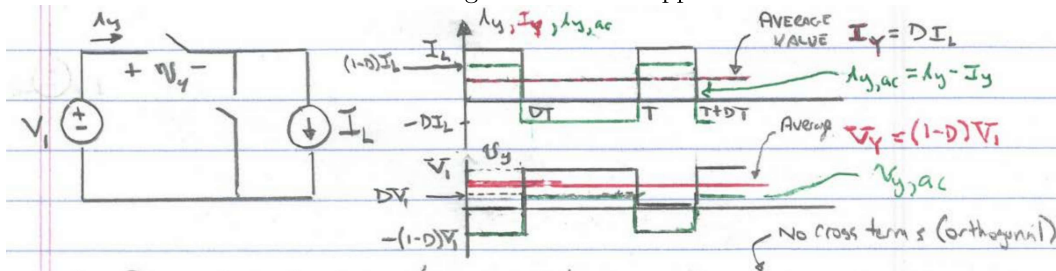
Assuming Ls and Cs are very big: $v_0(t) = V_0, i_2(t) = I_2$

Using average relation in P.S.S, $\langle v_L \rangle = \langle v_x \rangle - \langle v_2 \rangle$

$$\langle v_L \rangle = \frac{1}{T} [DT(v_1 - v_2) + (1 - D)T(-v_2)] = 0 \Rightarrow V_2 = Dv_1$$

2.1 Aside

What do switch 1 and switch 2 do? Let's ignore inductor ripple.



Average power into switch 1:

$$\langle P_1 \rangle = \langle (V_Y + v_{Y,AC})(I_Y + i_{Y,AC}) \rangle = \langle V_Y I_Y \rangle + \langle v_{Y,AC} i_{Y,AC} \rangle$$

Where the right side has no cross terms (orthogonal)

$$\begin{aligned} \langle P_1 \rangle &= D(1-D)I_L V_1 + \{D[-(1-D)^2 I_L V_1] + (1-D)[-D^2 I_L V_1]\} \\ &= \textcolor{red}{D(1-D)I_L V_1} - \textcolor{blue}{D(1-D)I_L V_1} = 0 \end{aligned}$$

Where the first term is average power into switch due to i_{Lv} and the second term is average power into switch due to $i_{ac,L}, v$ (Could not exactly read this part from original notes)

Switch S_1 takes average power in from the current, voltage and puts equal power out at ac current, voltage. Converts power (efficiently) from dc to ac waveforms! ("inverting" "switch") S_2 does the opposite (converts power from ac waveforms to dc waveforms!) (rectifying switch)

3 Review

Consider input current:

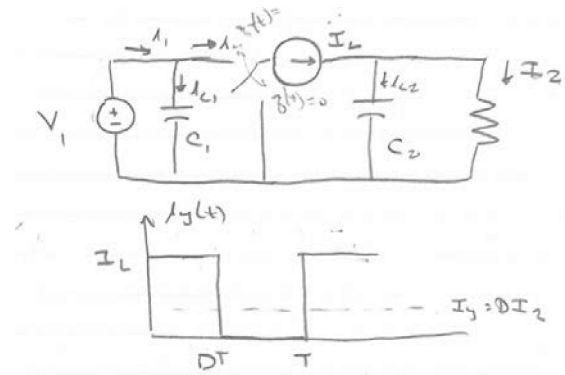
L's, C's big, $i_L(t) \approx I_L$

P.S.S. $\langle i_{c2} \rangle = 0 \therefore I_2 = I_L$

P.S.S.

$$\langle I_{c1} \rangle = 0 \therefore I_1 = \langle i_1 \rangle = \langle i_y \rangle = DI_2$$

$$\therefore I_1 = DI_2$$

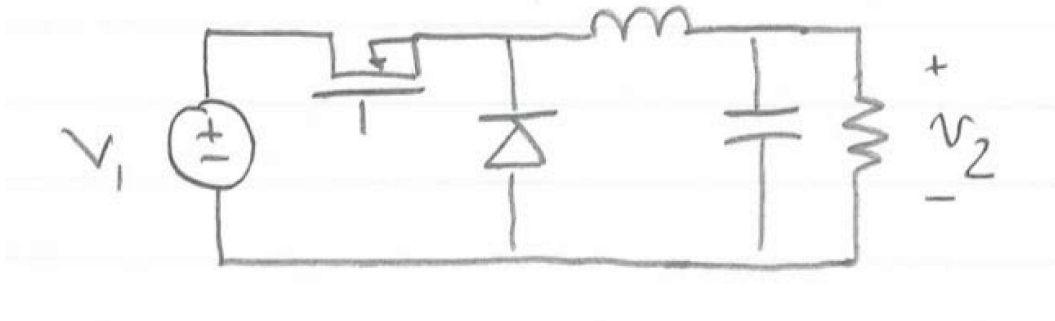


Combining with previous result that $DV_1 = V_2$: $I_1V_1 = I_2V_2 \leftarrow$ Lossless system!

Note: The trick is to be careful about when one is dealing with instantaneous variables and when one is dealing with average variables!

e.g. at a given instant, $i_y(t) \neq \langle i_y(t) \rangle$

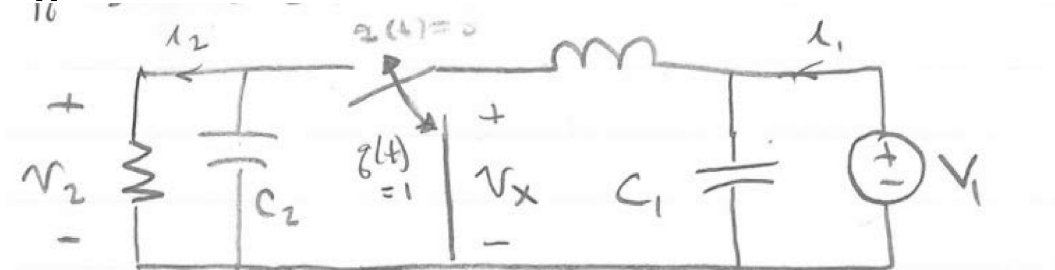
Switch implementation for this case, $v_1, v_2 > 0$



Power flows from 1 to 2.

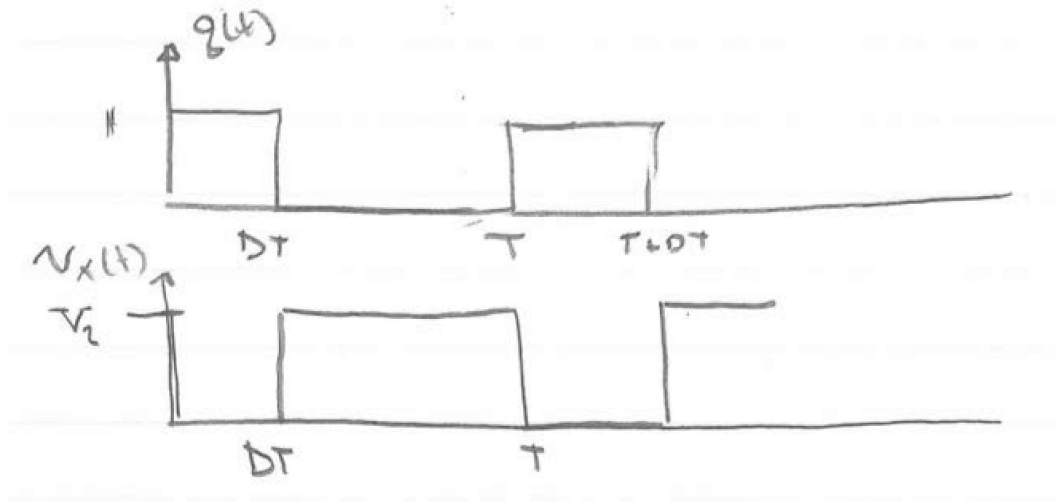
"Down" or "buck" converter. A type of "direct" converter because in one switch state, power flows directly from input to output.

Suppose we switch source and resistor.



*note: redefine $q(t) = 1$ as switch "down" position.

If C's, L's big, same analysis:

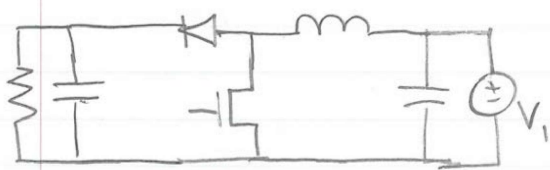


$$\langle v_L \rangle = 0 \therefore \langle v_x \rangle = (1 - D)V_2 = V_1$$

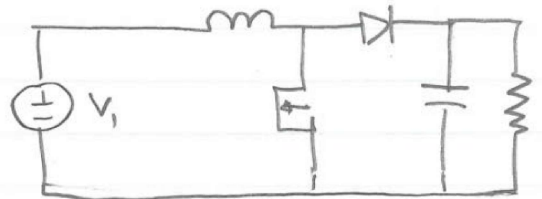
$$\therefore V_2 = \frac{V_1}{1 - D} \text{ and } \frac{I_2}{1 - D} = I_1$$

If $v_1, v_2 > 0$, then power flows from $2 \leftarrow 1$ and $v_2 > v_1$

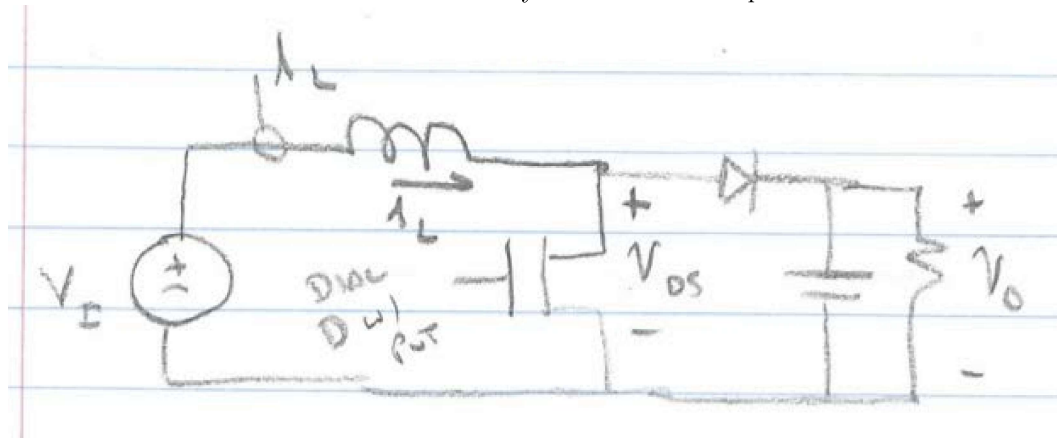
Boost converter (or "up" converter):



Sometimes drawn $L \rightarrow R$ power flow (But nothing fundamental about it).



* Show boost converter demo circuit built by Katie R. and Sauparna Das



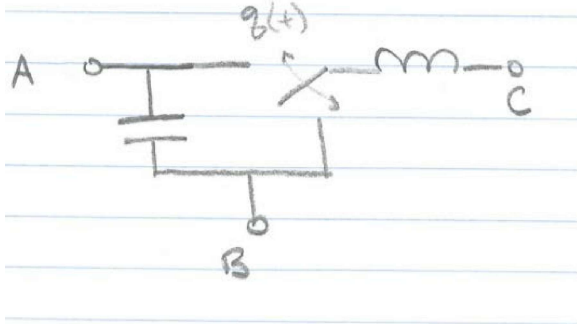
Show:

1. v_{DS}
2. i_L
3. v_0 on scope

Explain boost operation:

- Switch turns on, i_L rises and incrementally stores energy in L from V_I
- Switch turns off and this energy plus additional energy from V_I is transformed to output
- Steady state voltages are determined by $V_2 = \frac{V_1}{1-D}$

Either buck or boost can be viewed as a connection of a "canonical cell"



- Direct connection has B common
- one cannot tell power flow direction without knowing
 1. external networks
 2. switch implementation
 3. control

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